



An Analysis of Developer Skills and Future Directions

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OUTLINE



- Executive Summary
- Introduction
- Methodology
- Results
 - Visualization - Charts
 - Dashboard
- Discussion
 - Findings & Implications
- Conclusion
- Appendix

EXECUTIVE SUMMARY



- Data contextualization and analysis goal
- Methodology description
 - Data gathering
 - Data analysis
 - Data visualization
- Results shows the graphs and trend supported by various graphs
- Discussion of overall findings and implications regarding the results previously exposed
- Final conclusions of the findings in Dataset.

INTRODUCTION



- Stack Overflow's annual Developer Survey is the largest and most comprehensive survey of people who code around the world
- Results don't represent everyone in the developer community evenly
- Nearly 90,000 developers responded
- Trends
 - Technology worked with
 - Future technology to work

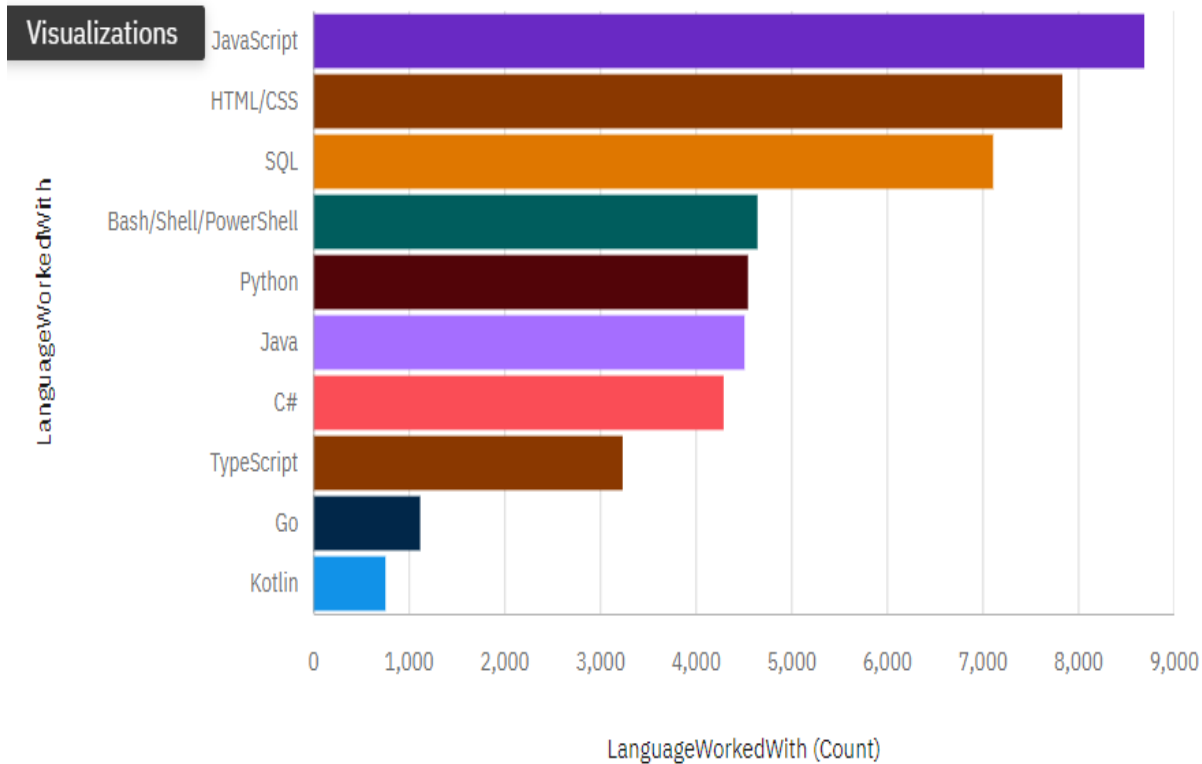
METHODOLOGY



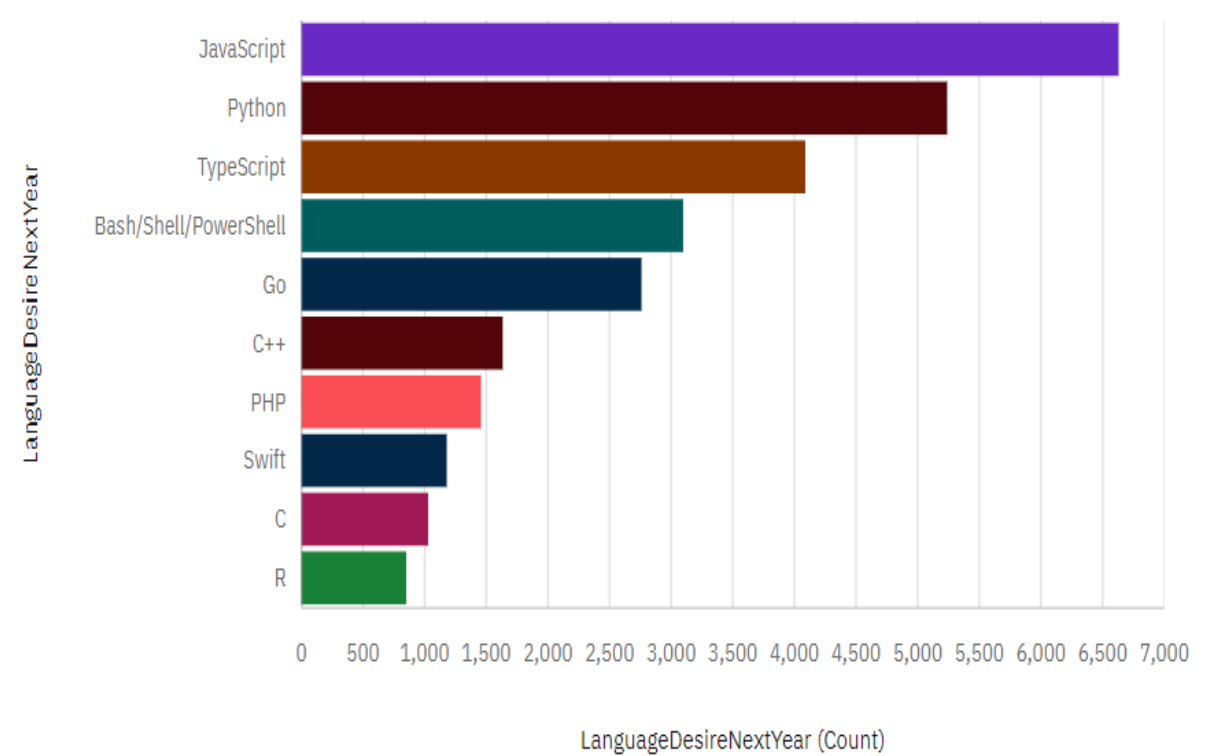
- Collect survey data
 - Web scraping
 - APIs
 - Request library
- Data wrangling
- Exploratory Data Analysis (EDA)
 - Distribution
 - Outlier Handling
 - Correlation
- Visualization
 - Distribution of data
 - Relationship
 - Comparison
 - Composition
- Dashboards

PROGRAMMING LANGUAGE TRENDS

Top 10 Language Worked with



Top 10 Language Desire Next Year



PROGRAMMING LANGUAGE TRENDS – FINDINGS & IMPLICATIONS

Findings

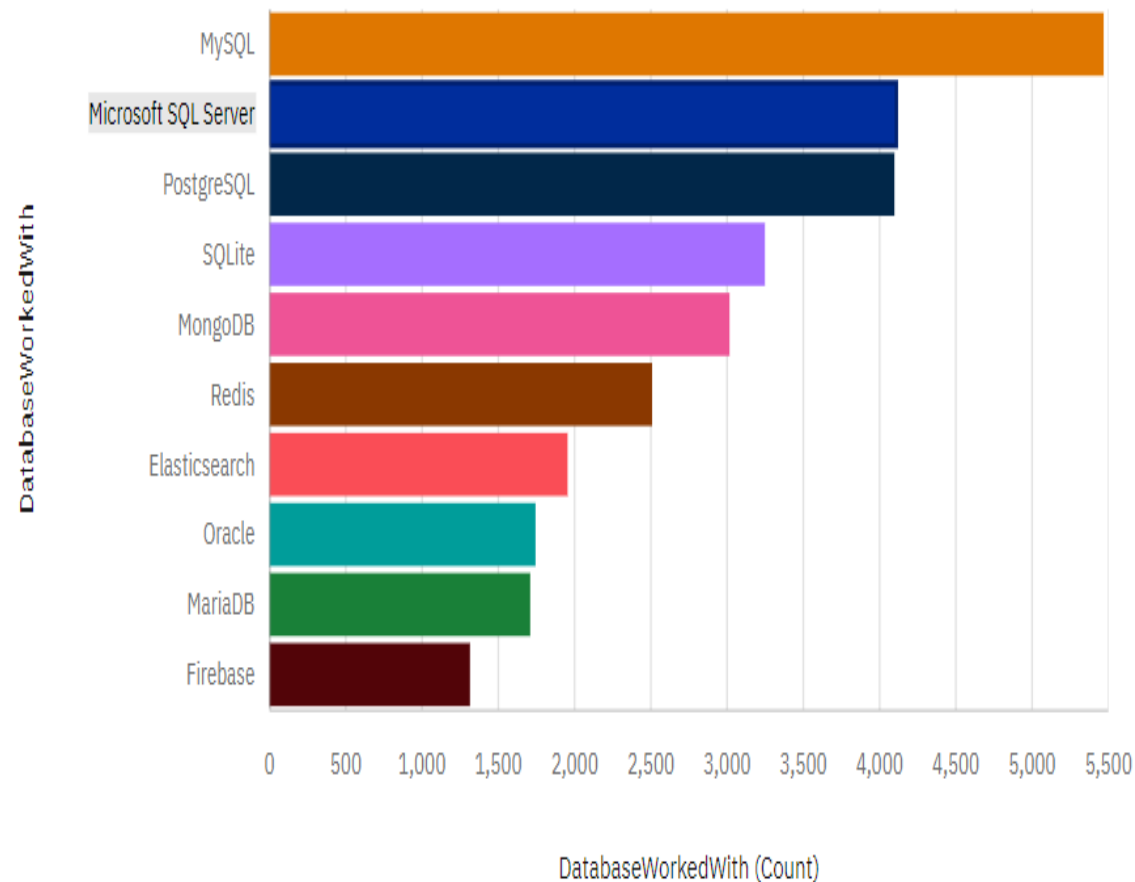
- Javascript is still the most widely used.
- HTML & CSS still in web development hence relevant.
- Python is desired due to the data science and backend usage

Implications

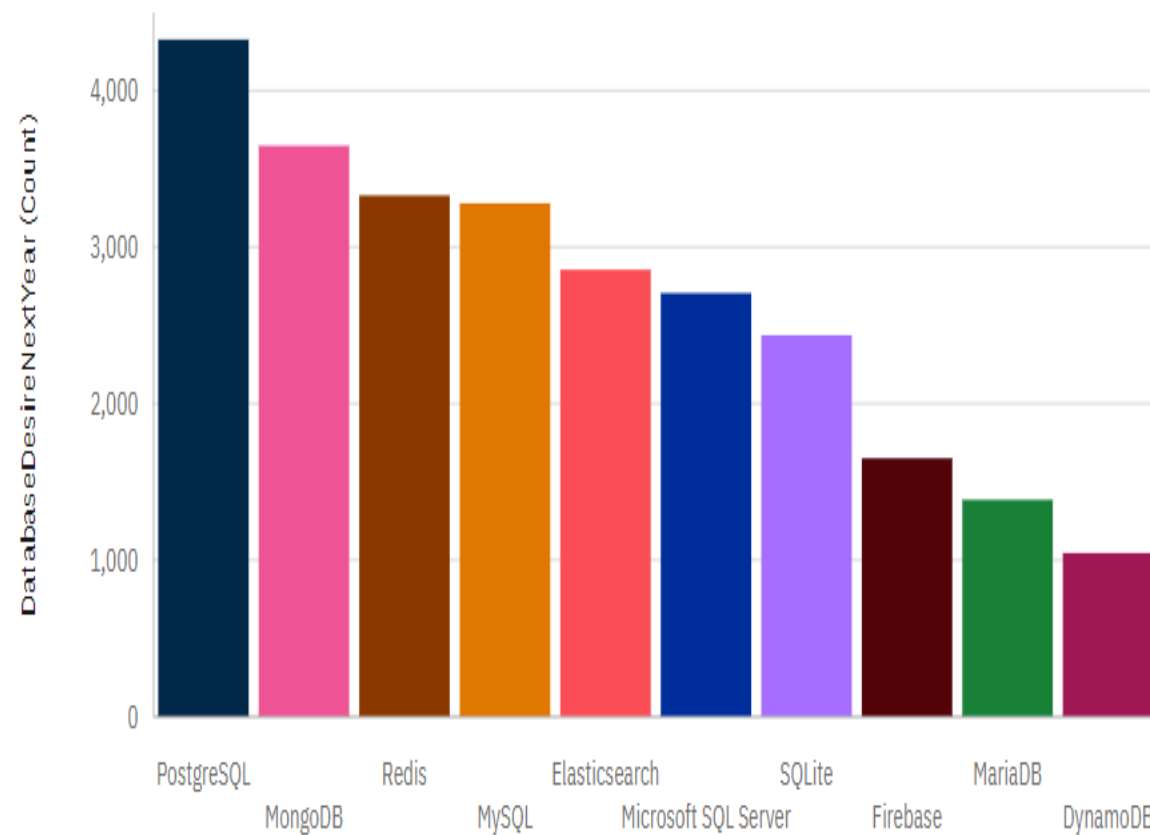
- Javascript and Python are the most desired hence consistence in market.
- Typescript, Go & rust show an emerging trend.

DATABASE TRENDS

Current Year



Next Year



DATABASE TRENDS – FINDINGS & IMPLICATIONS

Findings

- MySQL has the highest usage.
- PostgreSQL and Microsoft SQL Server have tight competition among databases.
- MongoDB and Redis are the most popular NoSQL Database
- Firebase has the lowest usage.

Implications

- PostgreSQL has a strong preference compared to the rest in the market.
- SQL still a fundament in the language for data handling
- Firebase has lost ground in the market.

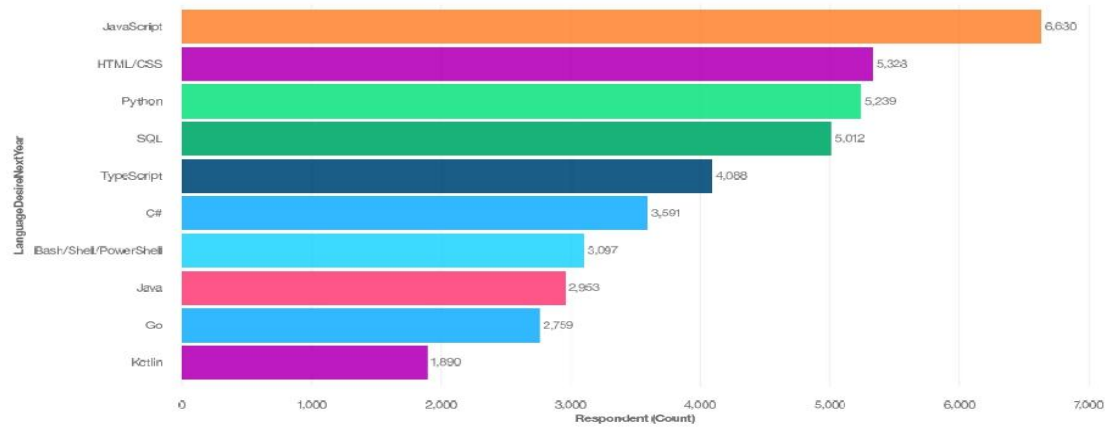
DASHBOARD



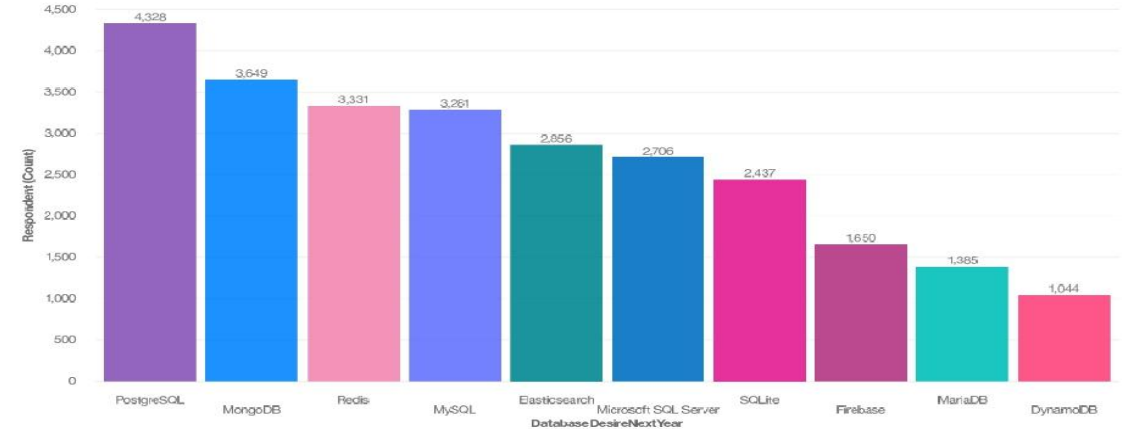
- Dashboard is created in IBM Cognos
- In the following three slides, dashboards created for different purposes as shown.

DASHBOARD 1: Current Technology Usage

Top 10 Languages Desired to Work With Next Year



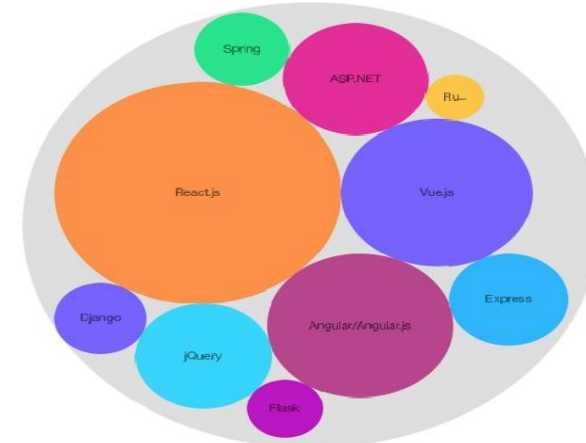
Top 10 Databases Desired to Work With Next Year



Platforms Desired to Work With Next Year



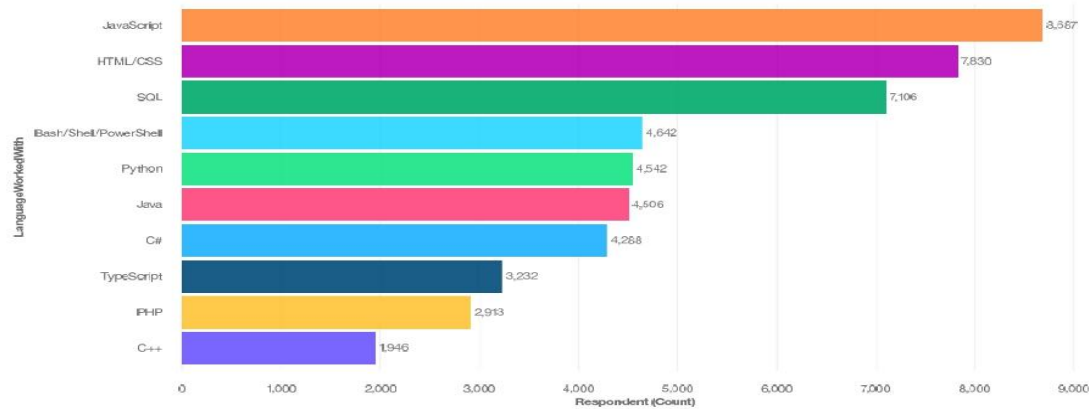
Top 10 Webframes Desired to Work With Next Year



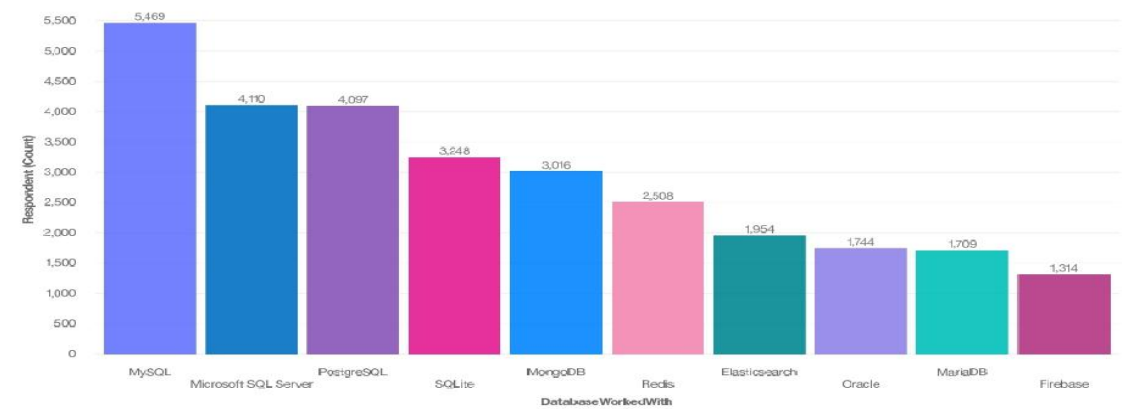
powered by IBM Cloud Pak for Data

DASHBOARD 2: Future Technology Usage

Top 10 Languages Worked With



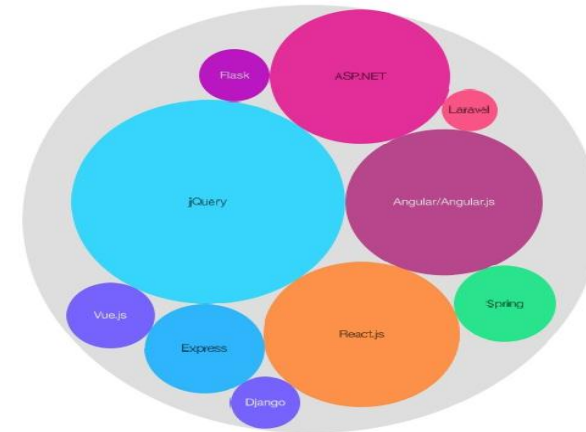
Top 10 Databases Worked With



Platforms Worked With



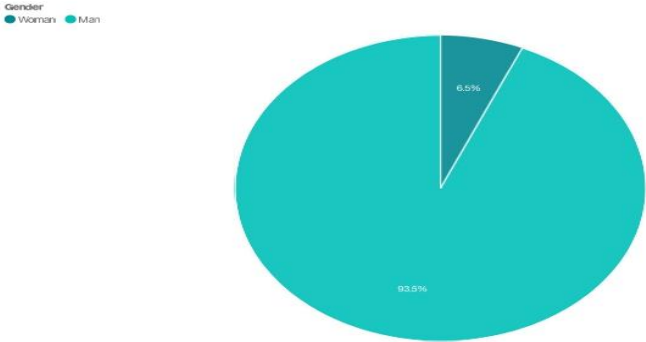
Top 10 Webframes Worked With



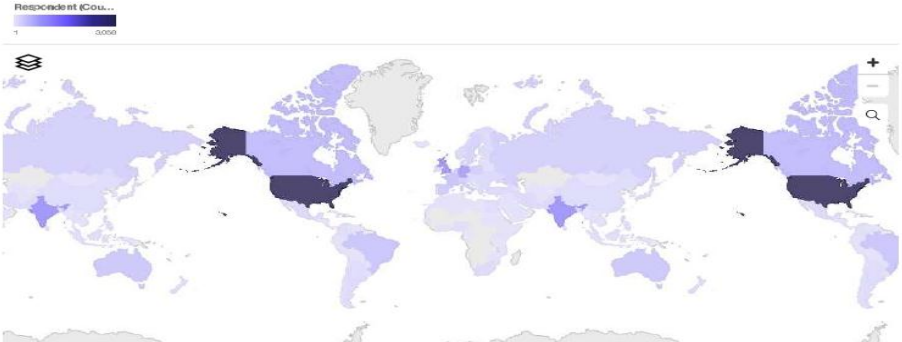
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DASHBOARD 3: Demographics

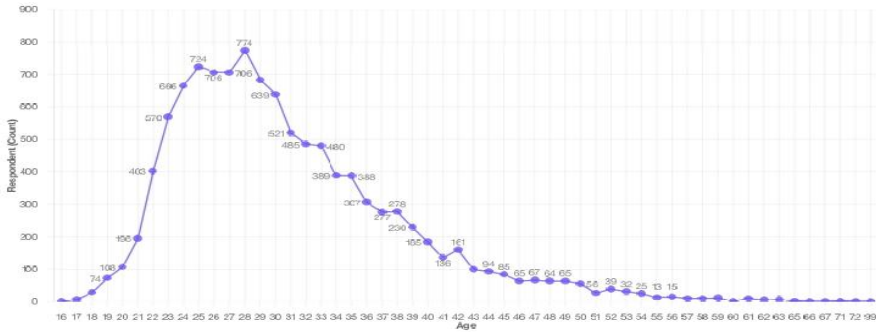
Percentage of Respondents based on Gender



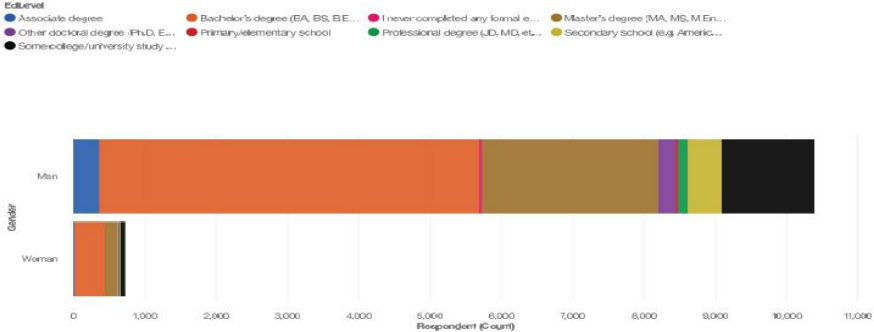
Respondents based on Location



Respondents based on Age



Respondents by Gender and Education Level



DISCUSSION

- 
- HTML, CSS, Javascript are the common languages across all developer age groups

OVERALL FINDINGS & IMPLICATIONS

Findings

- HTML, CSS, Javascript and SQL remain dominant in terms of usage by developers
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- JavaScript and Javascripts are the most desired languages and PostgreSQL among the databases for the next year.
- Over 90% young male developers.
- Developers mostly located in developed countries.

Implications

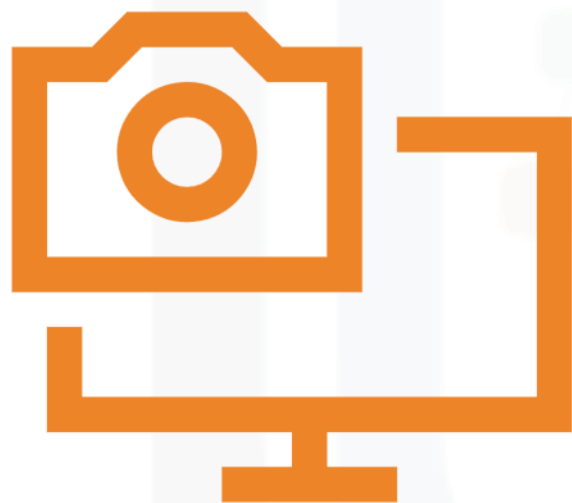
- TypeScript, Go and Rust web frameworks gaining followers.
- Global polarization of developers location and gender.
- Companies should hire professionals proficient in Python and PostgreSQL for future proofing their teams
- Companies should invest in training and upskilling for emerging technologies like TypeScript and Go and Redis to ensure their team stay relevant and competitive

CONCLUSION

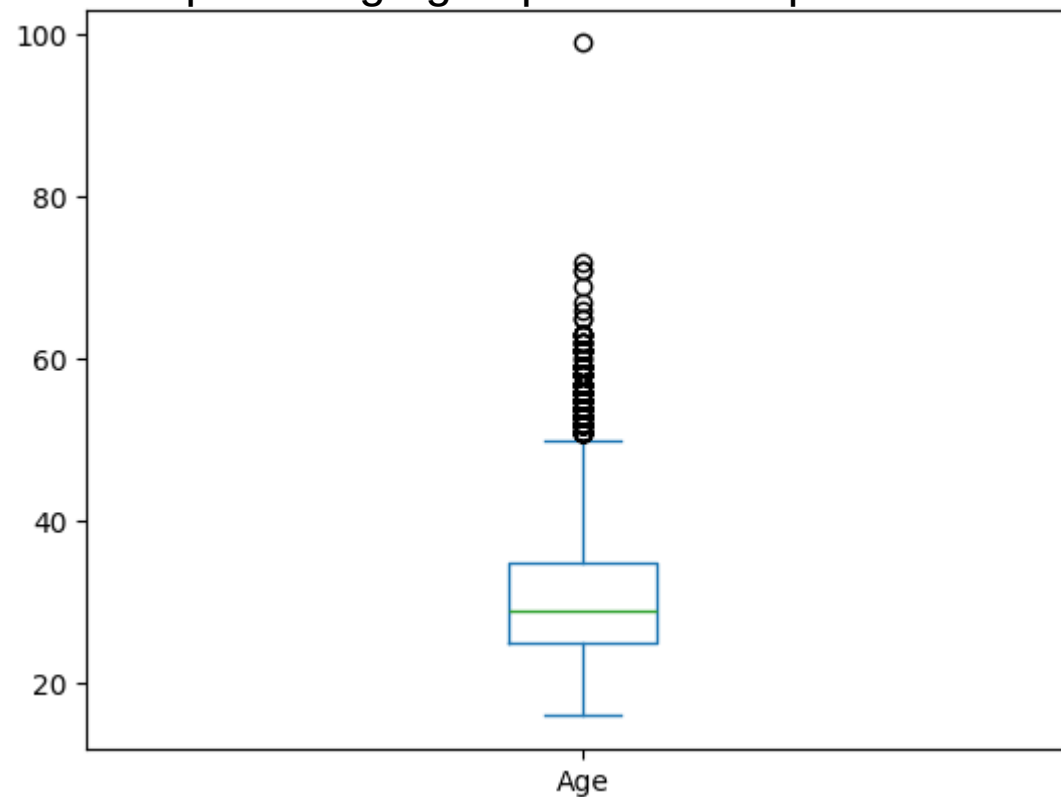


- The developer ecosystem is defined by the stability of foundational languages coupled with the constant growth of new , emerging technologies.
- Professional development must emphasize up skilling in high-demand technologies, particularly Python, TypeScript, and database like Postgresql
- A good idea of popularity trends of different tools, platforms and languages can be obtained.
- There is a job to be done to spread accessibility of this labor market to countries in development.
- Continuous learning and flexibility are a must have key career skill for successful career and rapid or fast revolving tech industry.

APPENDIX



Box plot of Age groups of Developer



POPULAR LANGUAGES

